

Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

Power



File: power.kicad_sch

FPGA_Power



File: fpga_power.kicad_sch

FPGA_Config



File: fpga_config.kicad_sch

JTAG



File: jtag.kicad_sch

Serial



File: serial.kicad_sch

USB



File: usb.kicad_sch

Connectivity



File: connectivity.kicad_sch

Input



File: voltage_input.kicad_sch

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague

Sheet: /
File: Controller_Board.kicad_sch

Title: Root

Size: A4 Date: 2026-03-10

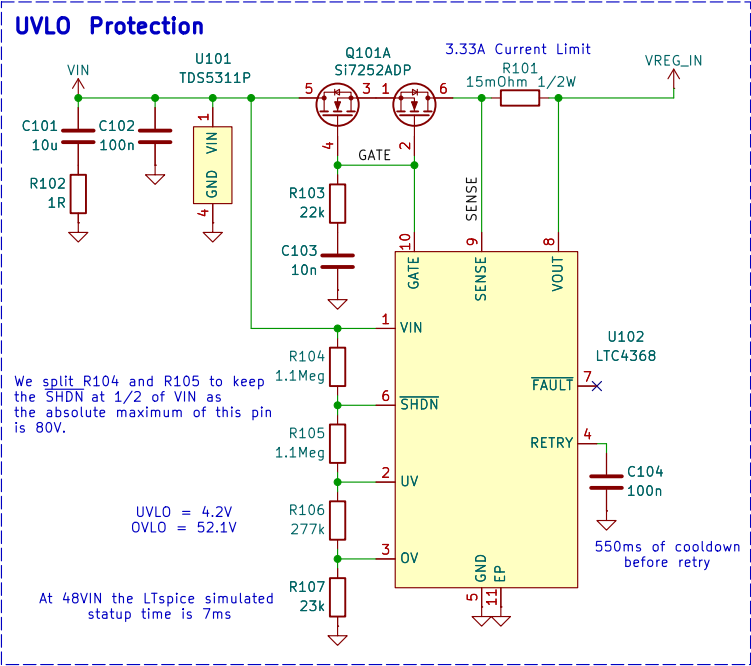
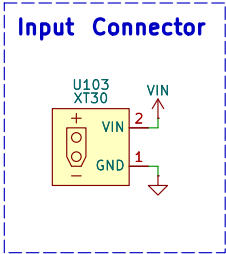
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Rev: 1.0

Id: 0/9

Input [1]

Maybe connect the P8_RX J4 and H4 and use the Bank 35 at 3V3



We split R104 and R105 to keep the SHDN at 1/2 of VIN as the absolute maximum of this pin is 80V.

UVLO = 4.2V
OVLO = 52.1V

At 48VIN the LTspice simulated statup time is 7ms

$$\text{In}[2]:= R3 = \frac{3 \times 10^{-3}}{10 \times 10^{-9}} * \left(\frac{4 - 0.5}{0.5} \right)$$

$$\text{Out}[2]= 2.1 \times 10^6$$

$$\text{In}[3]:= R1 = \frac{\left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) + 2.1 \times 10^6}{52} * 0.5$$

$$\text{Out}[3]= 23076.9$$

$$\text{In}[4]:= R2 = \left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) - 23076.9$$

$$\text{Out}[4]= 276923.$$

$$\text{In}[14]:= \text{NSolve} \left[2.2 \times 10^6 = \frac{3 \times 10^{-3}}{10 \times 10^{-9}} * \left(\frac{\text{UV} - 0.5}{0.5} \right), \text{UV} \right]$$

$$\text{Out}[14]= \{ \{ \text{UV} \rightarrow 4.16667 \} \}$$

$$\text{In}[15]:= \text{NSolve} \left[24000 = \frac{\left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) + 2.2 \times 10^6}{\text{OV}} * 0.5, \text{OV} \right]$$

$$\text{Out}[15]= \{ \{ \text{OV} \rightarrow 52.0833 \} \}$$

UV/OV Calculation

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Sheet: /Input/
File: voltage_input.kicad_sch

Title: Input

Size: A4 Date: 2026-03-27

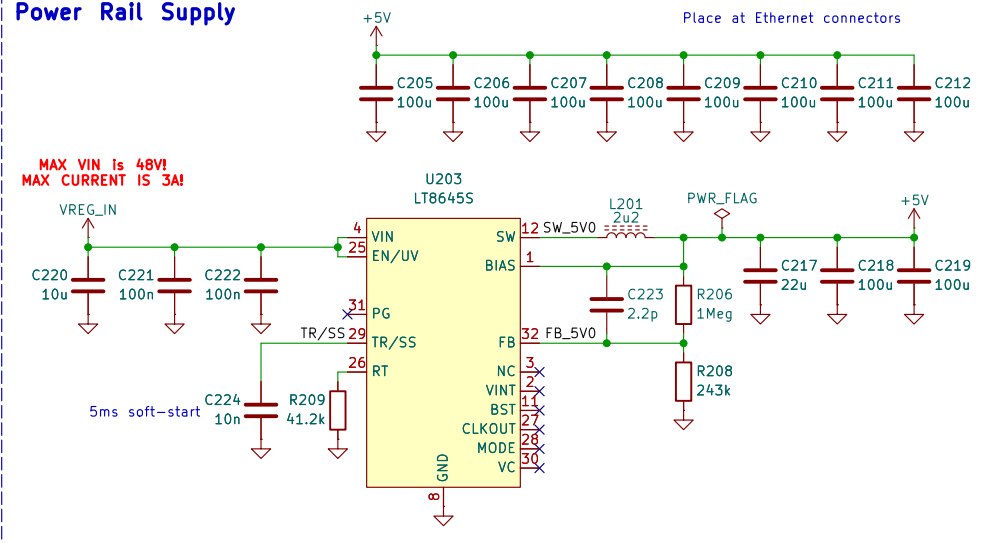
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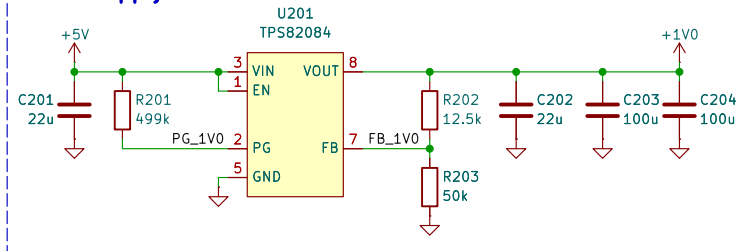
Id: 1/9

Power [2]

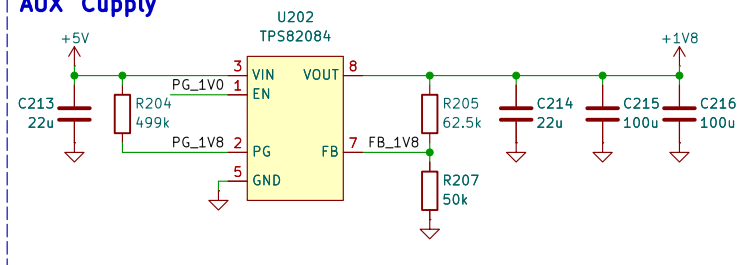
Power Rail Supply



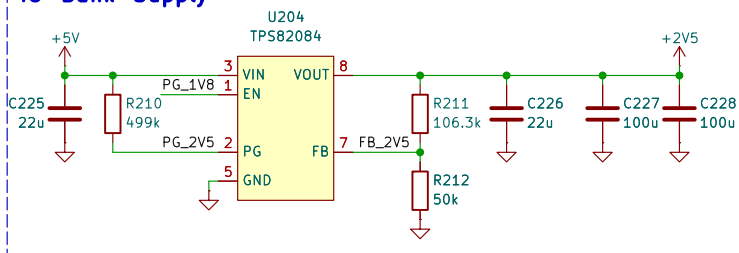
Core Supply



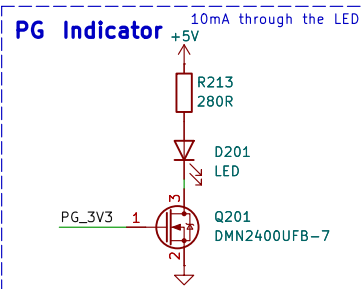
AUX Supply



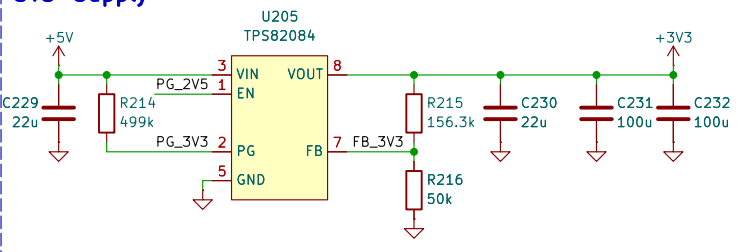
IO Bank Supply



PG Indicator



3V3 Supply



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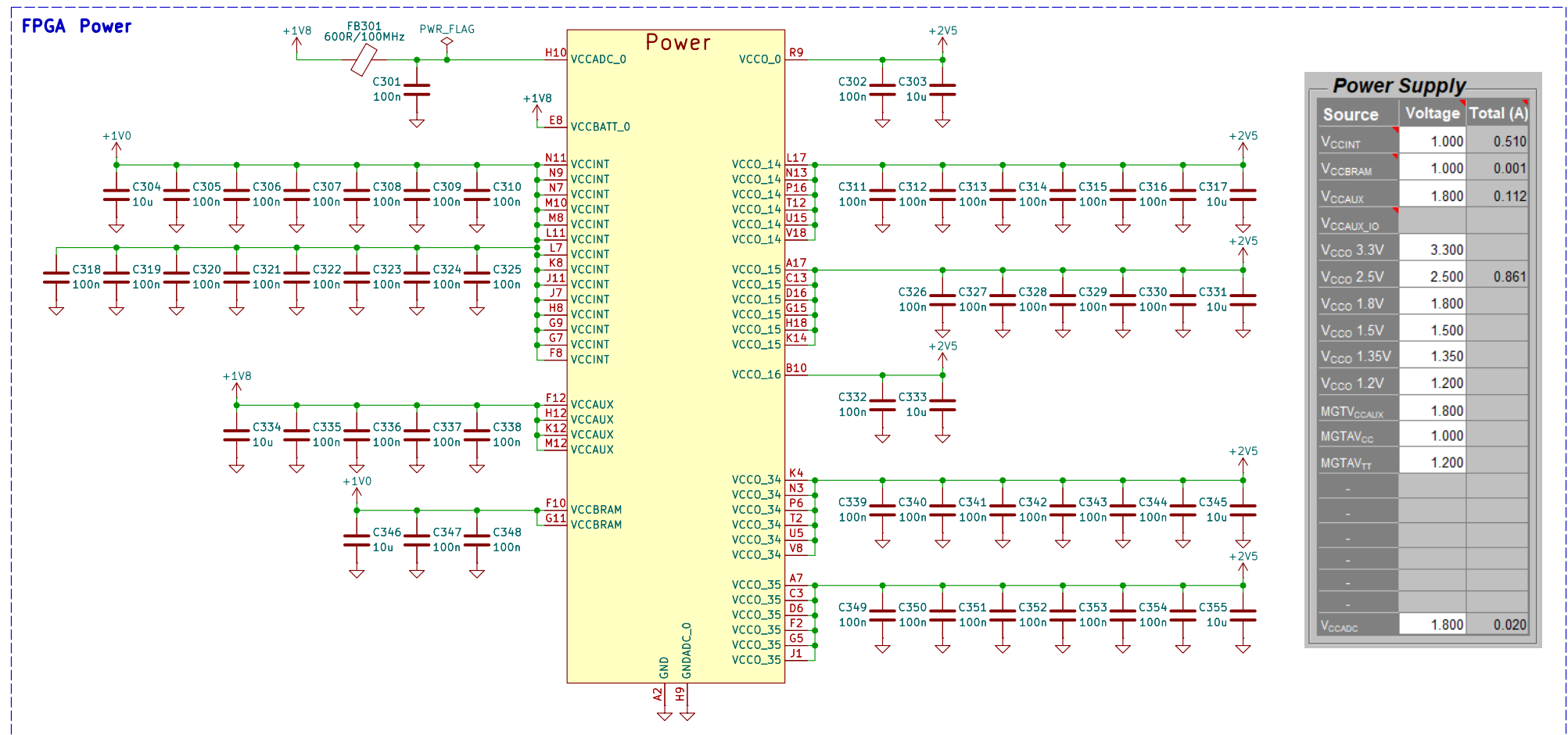
Sheet: /Power/
 File: power.kicad_sch

Title: Power

Size: A4 Date: 2026-03-07
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Rev: 1.0
 Id: 2/9

FPGA Power [3]



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Sheet: /FPGA_Power/
File: fpga_power.kicad_sch

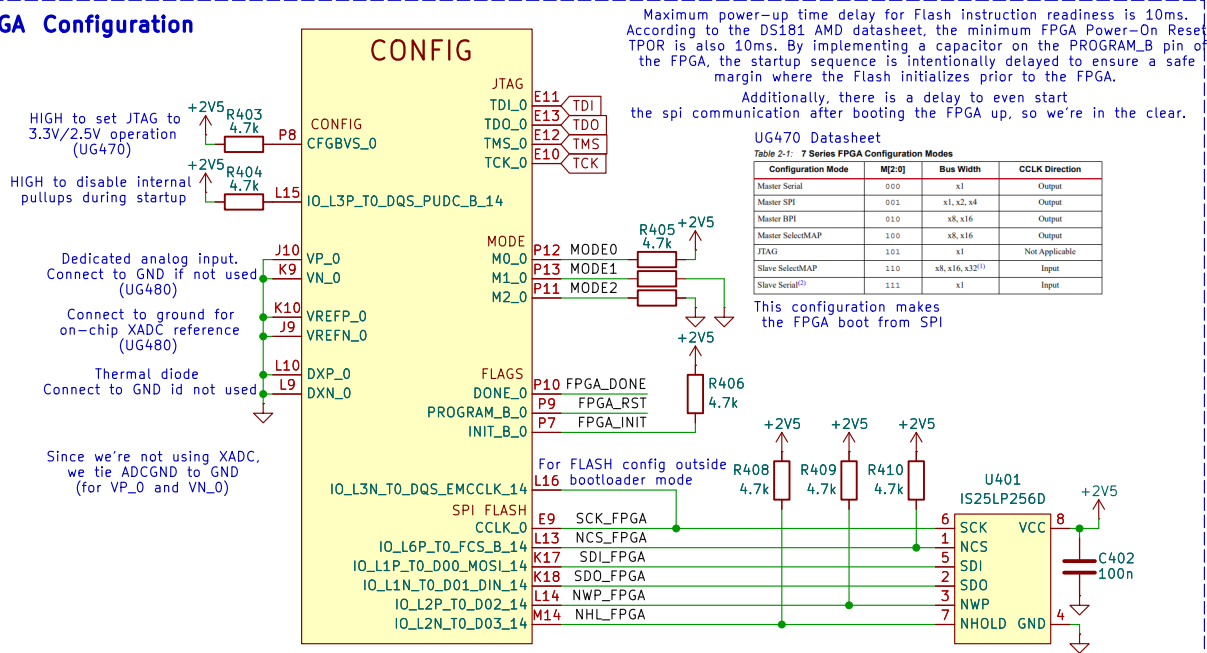
Title: FPGA Power

Size: A4 Date: 2026-03-10
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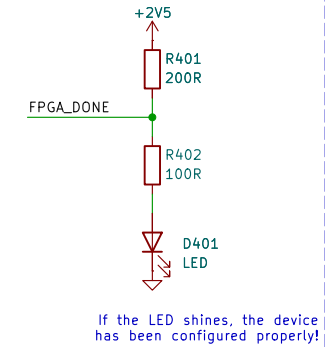
Rev: 1.0
Id: 3/9

FPGA Config [4]

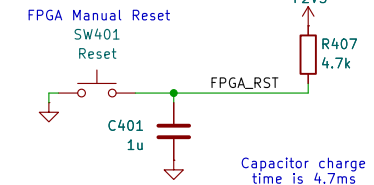
FPGA Configuration



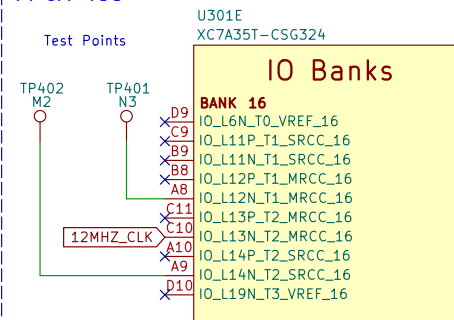
Configuration DONE



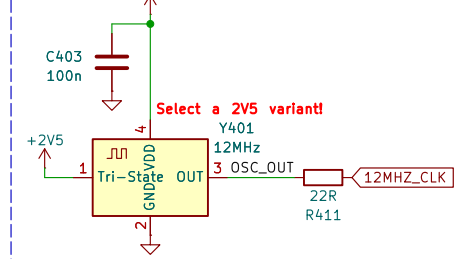
FPGA Manual Reset



FPGA IOs



External Oscillator



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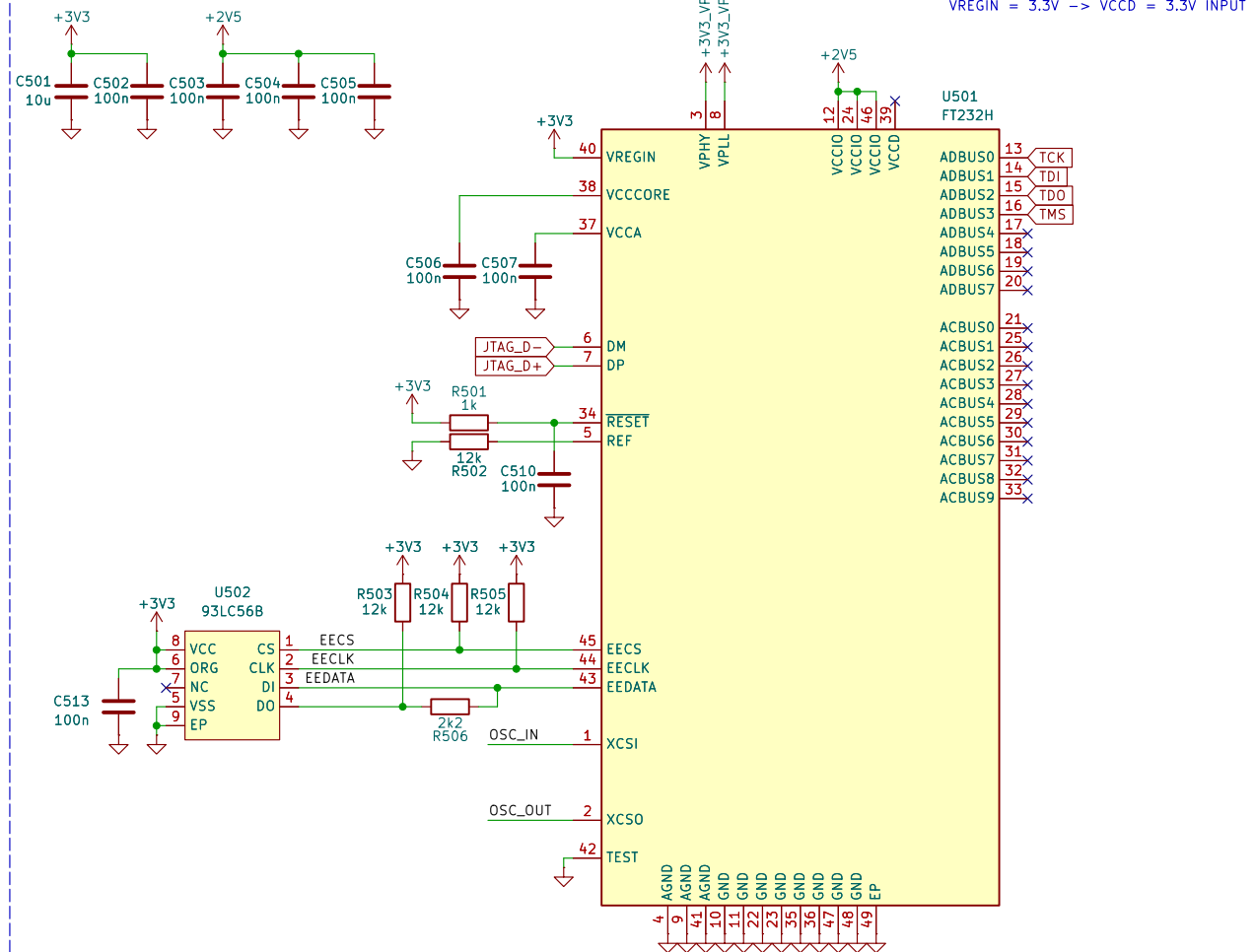
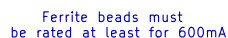
Sheet: /FPGA_Config/
File: fpga_config.kicad_sch

Title: FPGA Config

Size: A4
Date: 2026-03-10
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Rev: 1.0
Id: 4/9

D

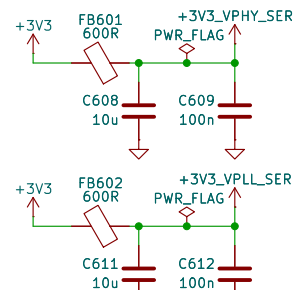


Id: 5/9

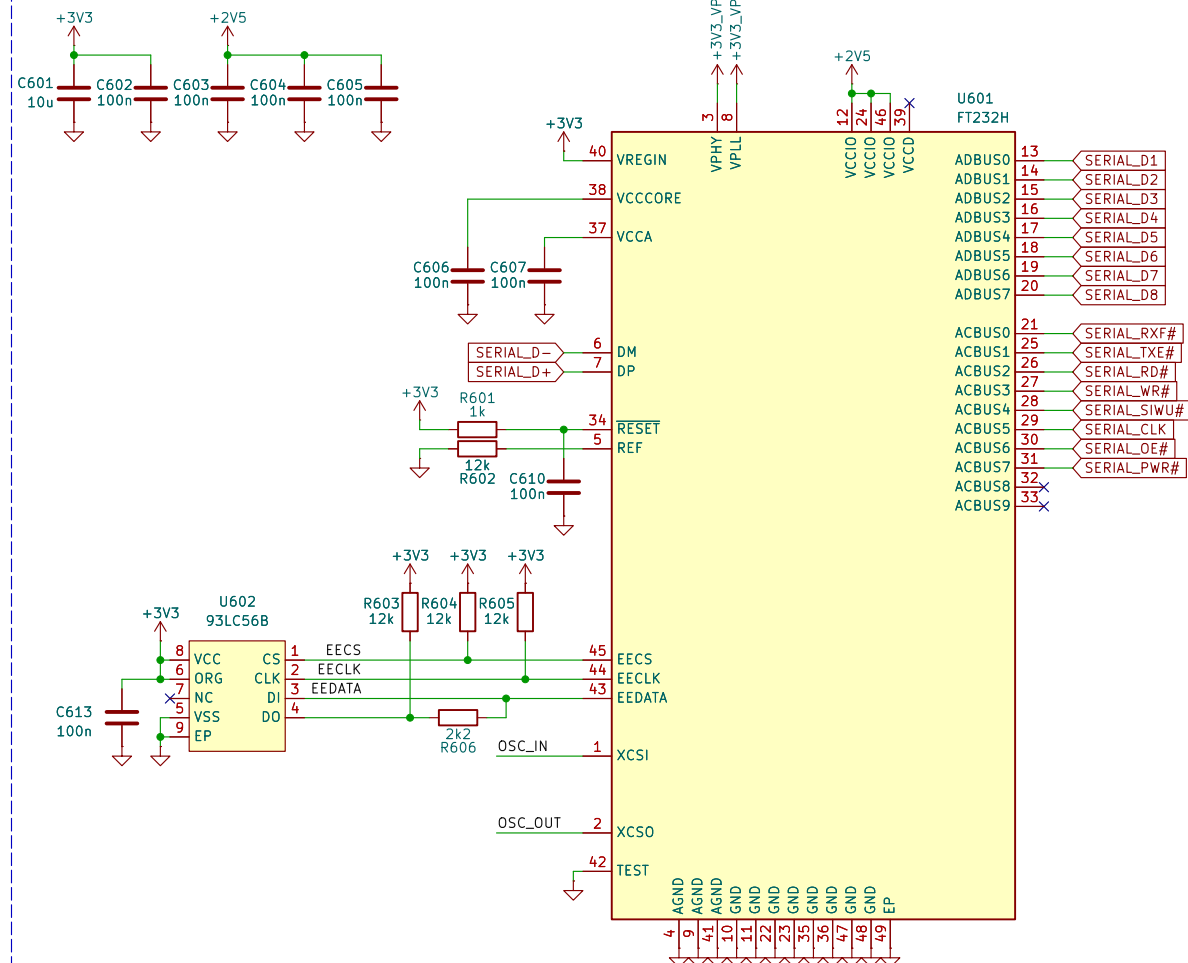
Serial [6]

FTDI USB-to-UART/SYNC FIFO

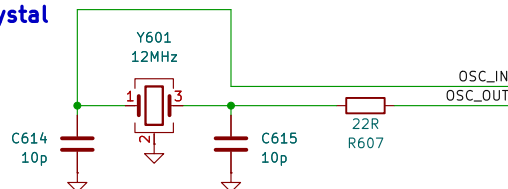
Filtering



Ferrite beads must be rated at least for 600mA



Crystal



$(10\text{pF Load Capacitance} - 5\text{pF Stray Capacitance}) * 2 = 10\text{pF Load Capacitors}$

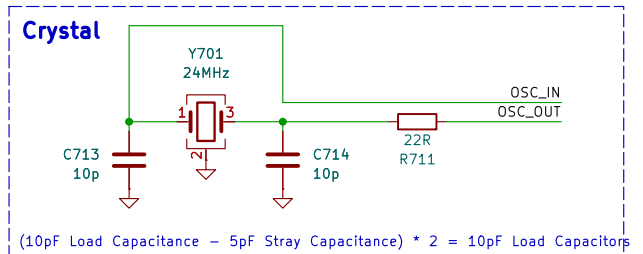
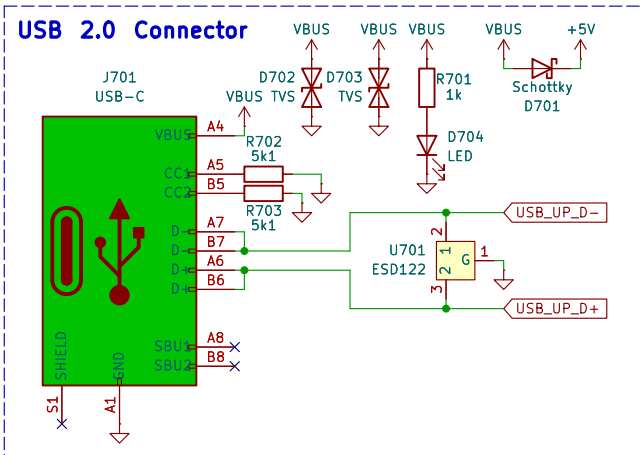
Reviewer: Michal Varsanyi
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Sheet: /Serial/
File: serial.kicad_sch

Title: Serial

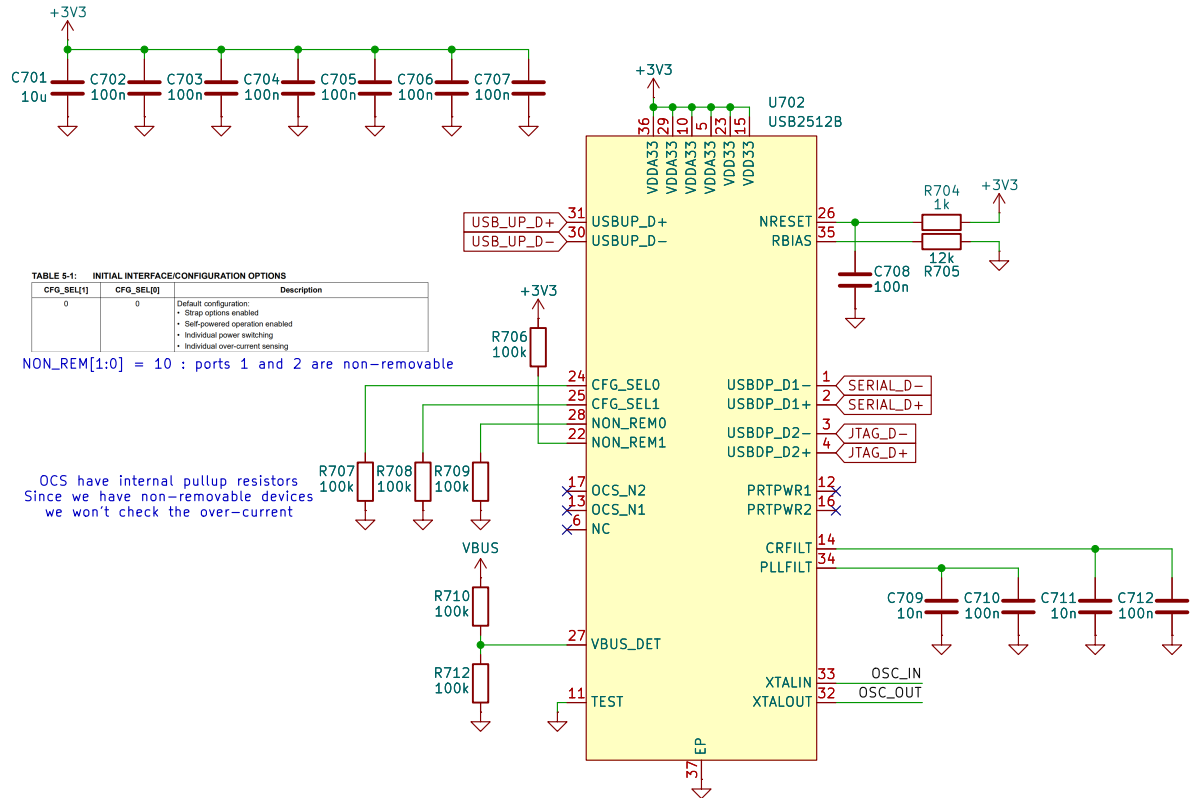
Size: A4 Date: 2026-03-11
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Rev: 1.0
Id: 6/9

USB [7]



USB 2.0 Hub



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Sheet: /USB/
File: usb.kicad_sch

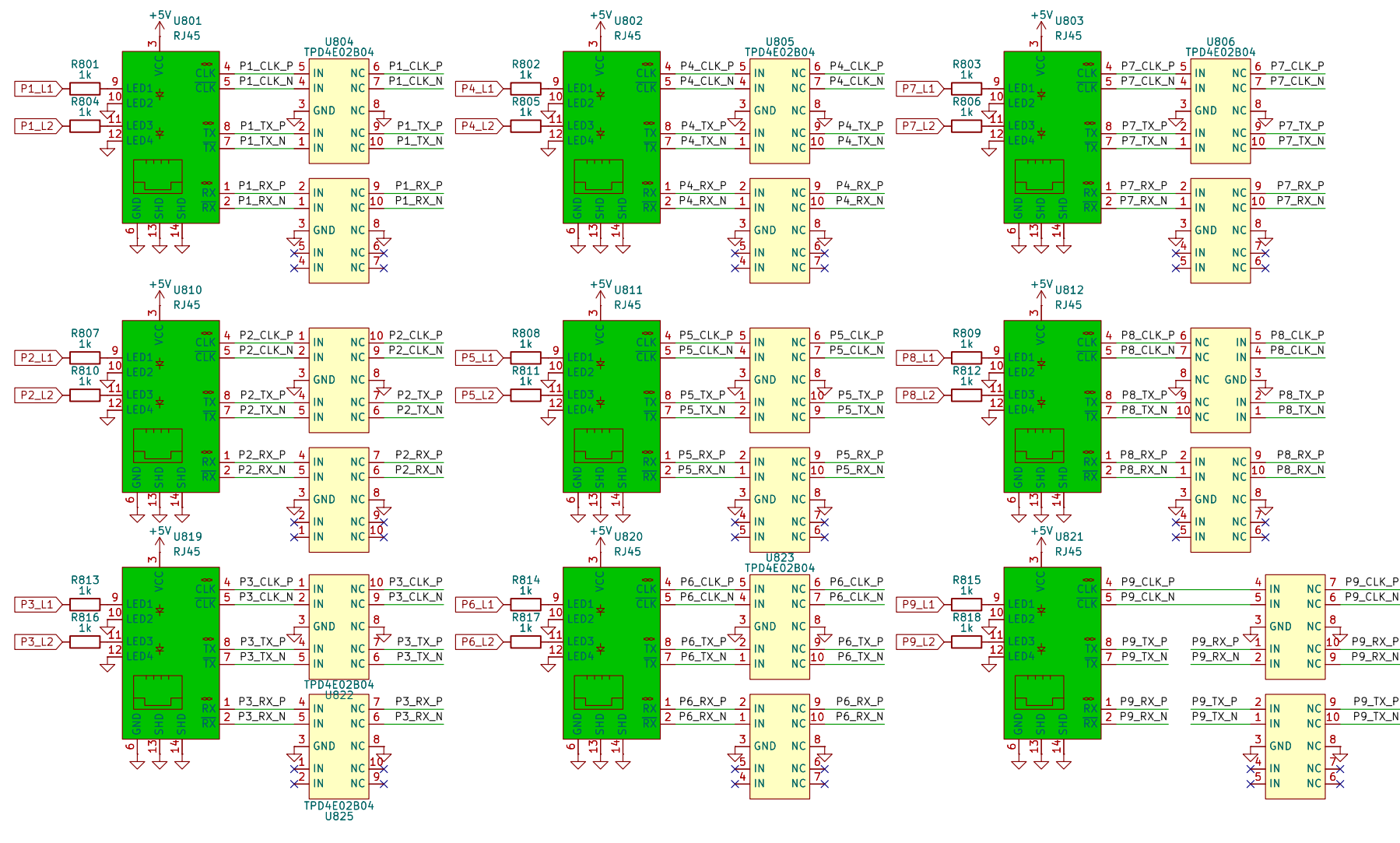
Title:

Size: A4	Date:
KiCad E.D.A. 10.0.0	

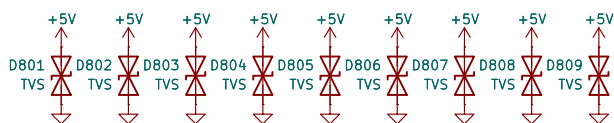
Rev: 1.0
Id: 7/9

Connectivity [8]

Ethernet Connectors



ESD Protection



M3 Mounting Holes



Please excuse this mess, I'll try to make this pretty later

IO Banks

BANK 15	
D14	IO_L1P_T0_AD0P_15
C14	IO_L1N_T0_AD0N_15
B13	IO_L2P_T0_AD8P_15
B14	IO_L2N_T0_AD8N_15
F12	IO_L3P_T0_DQS_AD1P_15
B12	IO_L3N_T0_DQS_AD1N_15
B11	IO_L4P_T0_15
A11	IO_L4N_T0_15
F13	IO_L5P_T0_AD9P_15
F14	IO_L5N_T0_AD9N_15
F12	IO_L6P_T0_15
B12	IO_L6N_T0_VREF_15
B16	IO_L7P_T1_AD2P_15
B17	IO_L7N_T1_AD2N_15
A15	IO_L8P_T1_AD10P_15
A16	IO_L8N_T1_AD10N_15
A13	IO_L9P_T1_DQS_AD3P_15
A14	IO_L9N_T1_DQS_AD3N_15
B18	IO_L10P_T1_AD11P_15
A18	IO_L10N_T1_AD11N_15
F15	IO_L11P_T1_SRCC_15
F16	IO_L11N_T1_SRCC_15
F15	IO_L12P_T1_MRCC_15
F15	IO_L12N_T1_MRCC_15
H16	IO_L13P_T2_MRCC_15
G16	IO_L13N_T2_MRCC_15
F15	IO_L14P_T2_SRCC_15
F16	IO_L14N_T2_SRCC_15
G14	IO_L15P_T2_DQS_15
G14	IO_L15N_T2_DQS_ADV_B_15
F17	IO_L16P_T2_A28_15
D17	IO_L16N_T2_A27_15
K13	IO_L17P_T2_A26_15
K13	IO_L17N_T2_A25_15
H17	IO_L18P_T2_A24_15
G17	IO_L18N_T2_A23_15
J14	IO_L19P_T3_A22_15
H15	IO_L19N_T3_A21_VREF_15
C16	IO_L20P_T3_A20_15
C17	IO_L20N_T3_A19_15
E18	IO_L21P_T3_DQS_15
D18	IO_L21N_T3_DQS_A18_15
G18	IO_L22P_T3_A17_15
J18	IO_L22N_T3_A16_15
J17	IO_L23P_T3_F0E_B_15
J18	IO_L23N_T3_FWE_B_15
K15	IO_L24P_T3_R51_15
J15	IO_L24N_T3_R50_15
G13	IO_0_15
K16	IO_25_15

IO Banks

BANK 14	
L18	IO_L4P_T0_D04_14
M18	IO_L4N_T0_D05_14
K12	IO_L5P_T0_D06_14
K13	IO_L5N_T0_D07_14
R18	IO_L7P_T1_D09_14
T18	IO_L7N_T1_D10_14
N14	IO_L8P_T1_D11_14
P14	IO_L8N_T1_D12_14
N17	IO_L9P_T1_DQS_14
R18	IO_L9N_T1_DQS_D13_14
L16	IO_L10P_T1_D14_14
M17	IO_L10N_T1_D15_14
N15	IO_L11P_T1_SRCC_14
N16	IO_L11N_T1_SRCC_14
P17	IO_L12P_T1_MRCC_14
R17	IO_L12N_T1_MRCC_14
R15	IO_L13P_T2_MRCC_14
T14	IO_L14P_T2_SRCC_14
T15	IO_L14N_T2_SRCC_14
R16	IO_L15P_T2_DQS_RDWR_B_14
T16	IO_L15N_T2_DQS_DOUT_CS0_B_14
V15	IO_L16P_T2_CSI_B_14
V16	IO_L16N_T2_A15_D31_14
U17	IO_L17P_T2_A14_D30_14
U18	IO_L17N_T2_A13_D29_14
U16	IO_L18P_T2_A12_D28_14
V17	IO_L18N_T2_A11_D27_14
P2	IO_L19P_T3_A10_D26_14
U11	IO_L19N_T3_A09_D25_VREF_14
U12	IO_L20P_T3_A08_D24_14
V12	IO_L20N_T3_A07_D23_14
V10	IO_L21P_T3_DQS_14
V11	IO_L21N_T3_DQS_A06_D22_14
U14	IO_L22P_T3_A05_D21_14
V14	IO_L22N_T3_A04_D20_14
J13	IO_L23P_T3_A03_D19_14
V13	IO_L23N_T3_A02_D18_14
T9	IO_L24P_T3_A01_D17_14
T10	IO_L24N_T3_A00_D16_14
R11	IO_0_14
M13	IO_L6N_T0_D08_VREF_14
R10	IO_25_14

IO Banks

BANK 34	
L1	IO_L1P_T0_34
M1	IO_L1N_T0_34
K3	IO_L2P_T0_34
L3	IO_L2N_T0_34
N2	IO_L3P_T0_DQS_34
N1	IO_L3N_T0_DQS_34
M3	IO_L4P_T0_34
M2	IO_L4N_T0_34
K5	IO_L5P_T0_34
L4	IO_L5N_T0_34
L6	IO_L6P_T0_34
L5	IO_L6N_T0_VREF_34
U1	IO_L7P_T1_34
V1	IO_L7N_T1_34
U3	IO_L8P_T1_34
U3	IO_L8N_T1_34
V2	IO_L9P_T1_DQS_34
V2	IO_L9N_T1_DQS_34
V5	IO_L10P_T1_34
V4	IO_L10N_T1_34
R3	IO_L11P_T1_SRCC_34
T3	IO_L11N_T1_SRCC_34
T5	IO_L12P_T1_MRCC_34
T4	IO_L12N_T1_MRCC_34
N5	IO_L13P_T2_MRCC_34
P5	IO_L13N_T2_MRCC_34
P4	IO_L14P_T2_SRCC_34
P3	IO_L14N_T2_SRCC_34
P2	IO_L15P_T2_DQS_34
R2	IO_L15N_T2_DQS_34
M4	IO_L16P_T2_34
N4	IO_L16N_T2_34
R1	IO_L17P_T2_34
T1	IO_L17N_T2_34
M6	IO_L18P_T2_34
N6	IO_L18N_T2_34
R6	IO_L19P_T3_34
R5	IO_L19N_T3_VREF_34
V7	IO_L20P_T3_34
V6	IO_L20N_T3_34
U9	IO_L21P_T3_DQS_34
V9	IO_L21N_T3_DQS_34
U7	IO_L22P_T3_34
U6	IO_L22N_T3_34
R7	IO_L23P_T3_34
T6	IO_L23N_T3_34
R8	IO_L24P_T3_34
T8	IO_L24N_T3_34
K6	IO_0_34
U8	IO_25_34

Reviewer: Michal Varsanyi
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Sheet: /Connectivity/
File: connectivity.kicad_sch

Title: Connectivity

Size: A3 Date: 2026-03-12
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Rev: 1.0
Id: 8/9